School of Computer Science and Engineering

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AI-Powered GitHub Repository Explainer Using Model Context Protocol

CS3240 – Foundations of Gen AI

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Contents

1. Title and Problem definition
2. Limitations of Existing Systems
3. Objectives
4. Methodology - Architecture Diagram
5. Innovations
6. Results
7. References

Problem Definition

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Developers often spend a lot of time understanding project structures, tracing dependencies, and interpreting poorly documented code, which slows down productivity. At the same time, Large Language Models (LLMs) face limitations in processing entire code repositories due to context size constraints. As a result, there is currently no unified system that effectively combines deep code understanding, semantic search capabilities, and clear explanations, making it harder to efficiently analyze and work with large and complex codebases.

Limitations of Existing Systems

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- Static analyzers focus only on syntax errors and cannot understand code meaning or behavior.
- Documentation tools require manual updates and often become outdated or incomplete.
- Code search tools are keyword-based and fail to capture semantic meaning.
- Large Language Models (LLMs) have limited context windows, so they cannot analyze entire large repositories.
- There is no unified system which interacts with live GitHub data using integrated search, retrieval, and explanation.

Objectives

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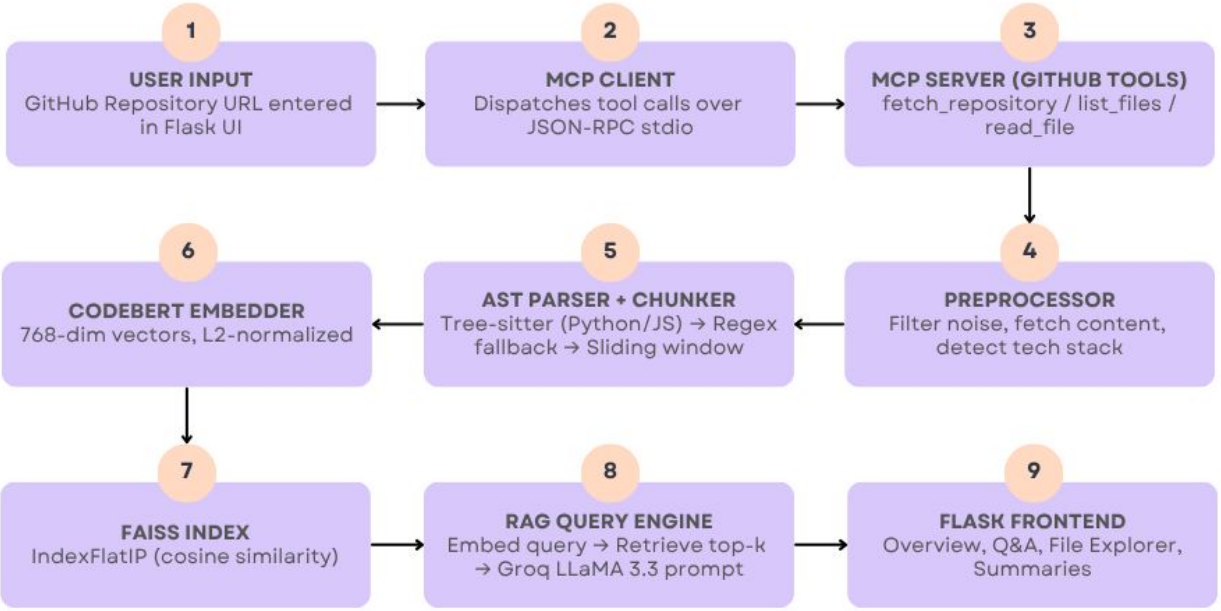
- The system should be able to fetch live data, manage large project files and only process the needed files using MCP
- Use Tree-sitter (AST) and regex fallback for code understanding
- CodeBERT to embed the parsed documents and FAISS for indexing
- Implement RAG pipeline for question-answering
- Provide concise overview of the repository

Architecture diagram



Innovations

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- We integrated MCP with RAG, which interacts with tools and retrieves content
- A combination of AST parsing and embeddings help understand code structure and meaning
- Structural analysis helps to understand how code is organized.
- Key innovation is repository-level explanation instead of single files
- The model can call other tools using modular tool-based LLM interaction

Results

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The system was evaluated across four open-source repositories of varying size and language, with each repository queried using five standard questions covering project purpose, entry point, authentication, API routes, and module connectivity.

- RAG retrieval was accurate, and it correctly traced code flow using only retrieved context.
- For unsupported languages, regex still gave useful structural explanations.
- Small repos were processed quickly, but large ones took time, but overall processing cost remained acceptable.
- Small datasets caused repetitive answers due to limited unique chunks.
- Caching avoided re-embedding for repeated queries, thereby reducing time.

References

[1] Z. Feng, D. Guo, D. Tang, N. Duan, X. Feng, M. Gong, L. Shou, B. Qin, T. Liu, D. Jiang, and M. Zhou, "CodeBERT: A Pre-Trained Model for Programming and Natural Languages," in Findings of EMNLP, 2020, pp. 1536–1547.
[2] W. Ahmad, S. Chakraborty, B. Ray, and K.-W. Chang, "A Transformer-based Approach for Source Code Summarization," in Proceedings of ACL, 2020, pp. 4998–5007.
[3] Y. Shi, Z. Wang, Y. Zhang, and X. Luo, "CodeRAG: Finding Relevant and Necessary Knowledge for Retrieval-Augmented Repository-Level Code Completion," in Proceedings of ACL, 2025.
[4] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-T. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in Proceedings of NeurIPS, 2020, pp. 9459–9474.
[5] S. Chen, Y. Xu, and X. Liu, "HierarchyNet: Learning to Summarize Source Code with Heterogeneous Representations," IEEE Transactions on Software Engineering, 2023.
[6] A. Yadav, R. Mehta, and S. Nair, "Uncovering Code Insights: Leveraging GitHub Artifacts for Deeper Code Understanding," in Proceedings of MSR, 2024.
[7] S. Thota, P. Reddy, and K. Sharma, "AI Based Code Error Explainer using Gemini Model," in Proceedings of ICACITE, 2024.
[8] S. Ganguly, A. Gupta, and P. Tiwari, "ML-Augmented Automation for Recovering Links Between Pull-Requests and Issues on GitHub," in Proceedings of SANER, 2024.
[9] Anthropic, "Model Context Protocol Specification," Anthropic Technical Documentation, 2024. [Online]. Available: https://modelcontextprotocol.io
[10] R. Wang, P. Kapur, and M. Nagappan, "An Empirical Study of Developer Onboarding in Large Software Projects," IEEE Transactions on Software Engineering, vol. 48, no. 9, pp. 3412–3427, 2022

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